High-Frequency Feature Transfer for Multispectral Image Super-Resolution

Fan Zhao, Xue Wu, Wenda Zhao, *Member, IEEE*, Zhepu Zhang, and Haipeng Wang

***Abstract*— Low-resolution characteristics of multispectral images restrict their usability. Various approaches (e.g., single-image super-resolution reconstruction (SISR) and pansharpening method) have been proposed to enrich the spatial details of low-resolution multispectral images (LRMSs) to obtain high-resolution ones. While the pansharpening method inevitably depends on panchromatic (PAN) images, which limits its application scenarios, SISR does not need auxiliary images, yet the blurred edge details within the reconstructed super-resolution multispectral (SRMS) images still remain a big challenge. In this work, we propose a novel high-frequency feature transfer (HFFT-PAN) method for multispectral image super-resolution to tackle the above drawbacks. Specifically, we first exploit the inherent low-frequency features among LRMS images to facilitate the extraction of high-frequency features from PAN images. After that, the high-frequency features from PAN images are transferred to the reconstruction procedure of multispectral images so that the SRMS images can not only benefit from the edge detail information from PAN images but also avoid the utilization of any PAN images during inference. Moreover, we employ an additional contrastive loss during training to ensure the fidelity of the generated SRMS images. Qualitative and quantitative evaluations exhibit that the proposed method performs favorably against state-of-the-art methods. The model and code are available at https://github.com/wx0110wx/HFFT.**

***Index Terms*— Contrastive learning, high-frequency feature transfer, multispectral image super-resolution.**

I. Introduction

MULTISPECTRAL images are formed by capturing signals of multiple optical spectrum bands from sensors, leading to adequate spectral information. Because of this, multispectral images are widely adopted in various kinds of applications including ground object classification, object detection, and surface change detection. However, due to the hardware limitations of sensors, it is always hard to make a tradeoff between spectral resolution and spatial resolution [1], which, as a result, causes the loss of high-frequency information. That drawback of multispectral images significantly limits their utilization in various applications to a certain extent.

To improve the usability of low-resolution multispectral images (LRMSs), a large number of methods have been proposed to recover their edge details, which can be roughly divided into two categories: single-image super-resolution reconstruction [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] and pansharpening methods [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23]. Single-image super-resolution reconstruction (SISR) method aims to reconstruct a corresponding high-resolution version with clear high-frequency features from a given low-resolution image via a specific super-resolution algorithm, as shown in Fig. 1(a). The SISR method can be further divided into traditional SISR methods and deep learning-based ones. While traditional SISR methods [2], [3], [24], [25] mostly rely on old-school digital image processing techniques, deep learning-based methods [9], [12], [26], [27] aim to reconstruct super-resolution multispectral (SRMS) images by learning the nonlinear mapping between LRMS images and provided high-resolution ground-truth (GT) images. Compared with traditional methods, deep learning-based methods have an edge in recovering high-frequency information, including textures, edges, and so on. On the other hand, pansharpening approaches consider leveraging image fusion paradigms as solutions. By introducing related PAN images as supplementary inputs, the lack of high-frequency information in LRMS images can be efficiently acquired from PAN modality, thus obtaining SRMS images that combine both high-frequency and spectral information, as shown in Fig. 1(b). Pansharpening methods can also be further divided into component substitution-based methods, multiresolution analysis-based methods, degradation model-based methods, and deep learning-based methods. The component substitution-based methods [13], [14], [15], [16], [28] suggest replacing the spatial component of LRMS images with PAN images. Multiresolution analysis-based methods [17], [29], [30] employ multiscale decomposition technologies to separate the high-frequency information from PAN images, following with adaptively fusion with LRMS images. Among the degradation model-based methods [31], [32], [33], the pansharpening task is regarded as an image restoration problem, where LRMS images and PAN images are considered as the degradation results of SRMS images in the spatial domain and spectral domain, respectively. More recently, deep learning-based methods [18], [20], [22], [34] make full use of the strong feature extraction and fitting ability

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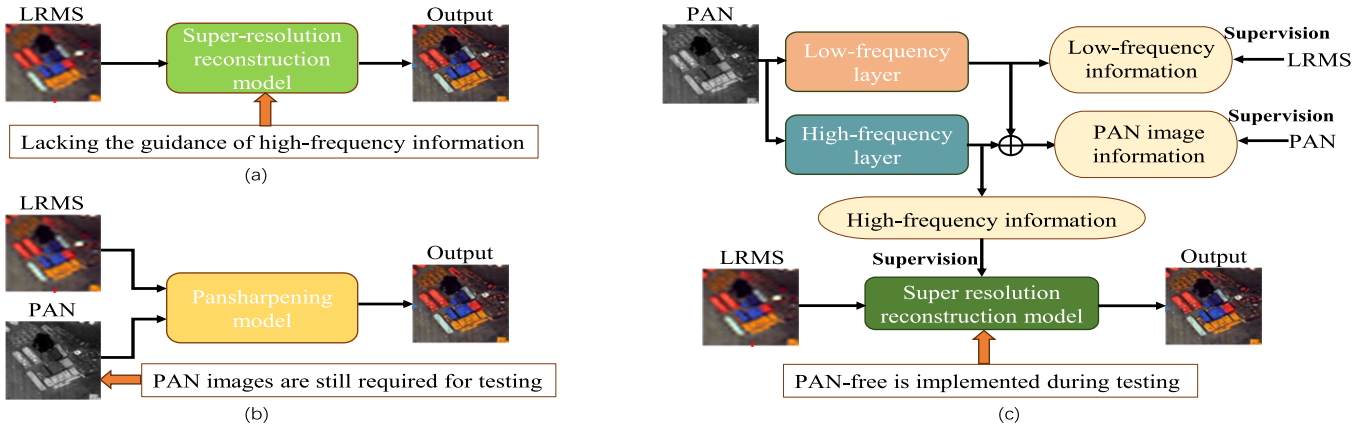


Fig. 1. Motivation of our proposed method. (a) SISR. Those approaches only require LRMS inputs, causing a lack of high-frequency information as guidance. (b) Pansharpening. High-frequency information is provided by PAN images, yet such methods inevitably demand PAN inputs for both training and testing. (c) Our method not only employs PAN images as a supplement while training but also implements PAN-free during testing.

of neural networks to extract features from both LRMS and PAN images layer by layer before fusing together, achieving more accurate sharpening results.

Both existing SISR and pansharpening methods have obvious drawbacks. SISR methods lack the guidance of high-frequency information, as shown in Fig. 1(a). Since high-frequency information is scanty in LRMS images, SISR methods struggle to reconstruct texture features and detail information that did not originally exist. Although the pansharpening methods manage to effectively improve the spatial resolution of LRMS images by fusing high-frequency information from PAN images, they inevitably demand PAN inputs during testing phases just like their training procedures, as shown in Fig. 1(b).

In addition, reference image-based SISR methods [35], [36], [37], [38] are proposed to provide high-frequency information guidance. They extract the high-frequency features from additional HR images and then design the spatial alignment and patch matching to align the features with the ones of LR images. However, the reference images usually do not match the LR images exactly, which results in difficult alignment and complex calculations. Moreover, the reference images are needed to input the model in the test stage. Therefore, we propose a novel high-frequency feature transfer (HFFT-PAN) method that takes advantage of the characteristics of the matched PAN images. HFFT-PAN uses PAN images as the reference to extract the aligned high-frequency features and then transfers these features to the LR images by knowledge distillation, thereby avoiding introducing PAN inputs during testing, as shown in Fig. 1(c).

Due to the lack of high-frequency components as supervision, we design a teacher network with a two-branch structure to separate the high-frequency information from PAN images first. Two parallel branches are expected to extract the low-frequency and high-frequency features from PAN inputs, respectively, both of which are supervised under a self-reconstruction manner. To ensure both branches can leverage their distinct functions effectively, we utilize the fact that the LRMS image contains similar low-frequency features with its corresponding PAN pair. By restricting the low-frequency branch to extract analogous low-frequency features with LRMS images, high-frequency information will be learned by the other branch spontaneously to guarantee the self-reconstruction result. After that, we transfer the high-frequency information extracted from PAN images to enhance the related feature recovery during the super-resolution procedure of LRMS images. Since all the auxiliary knowledge from PAN images acts as supervision to guide the super-resolution of LRMS during training, no PAN inputs are required during inference. To further improve the image quality of super-resolution results, a contrastive learning mechanism is employed additionally to reinforce the high-frequency part recovered by the network.

More specifically, the teacher network consists of two branches including a high-frequency layer (THF) and a low-frequency layer (TLF). The goal of the teacher network is to separate both the low- and high-frequency parts from the PAN inputs through TLF and THF, respectively, in a self-reconstruction manner. To make the separation precise, we introduce similar low-frequency information derived from LRMS inputs as supervision to the TLF branch. As a result, the THF branch is forced to learn the high-frequency features from PAN images to ensure the integrity of information. In order to realize the transfer of high-frequency features from PAN inputs to the process of LRMS images yet avoid using any PAN inputs while inference, we construct a super-resolution reconstruction network (SHF) that implements reconstruction of LRMS image under the supervision of features extracted by THF. By doing so, not only the SRMS image can benefit from the knowledge of PAN images but also no extra PAN inputs truly participated in the super-resolution paradigm. Moreover, we add a contrastive learning loss (CLM), which takes original LRMS images as negative samples and GT images as positive samples to constrain the super-resolution results. By minimizing the distance between generated SRMS images and positive samples while maximizing its distance with negative ones, the fidelity of the SRMS results is ensured.

To summarize, the main contributions of this article are threefold.

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1) We transfer the high-frequency features from PAN images to LRMS images by distillation, so as to take advantage of the rich edge details from PAN images during training while avoiding the usage of PAN images in the test phase.
2) We design a two-branch PAN image self-reconstruction network to extract high-frequency features. Multispectral images contain low-frequency features, which can be treated as supervision to help one branch extract the low-frequency components from the PAN image, thereby forcing the other branch to extract the high-frequency components of the PAN image.
3) Extensive experiments show that our proposed HFFT-PAN performs favorably against state-of-the-art methods both quantitatively and qualitatively.

II. Related Work

Deep learning techniques have been widely adopted to improve the details of multispectral images recently, which can be simply divided into two main categories: single-image super-resolution methods and pansharpening methods.

A. Single-Image Super-Resolution Methods

Various deep learning-based SISR algorithms have been investigated over the past decade, including convolutional neural network-based methods [8], [26], [39], residual network-based methods [5], [11], and recursive neural network-based methods [27], [40].

Dong et al. [26] introduce deep learning to SISR for the first time and propose the SRCNN model. SRCNN uses an interpolation algorithm to upsample the image and reconstructs the image through a three-layer convolutional neural network. Dong et al. [8] further propose the FSRCNN model. Compared with their previous SRCNN, both the computational efficiency and the reconstruction quality are significantly improved. Besides, the upsampling process of low-resolution input is integrated within the network, simplifying the preprocessing procedure. Zhang et al. [39] propose a scalable SISR model, namely, SRMD, in which blur kernel, noise, and low-resolution image are input to the network simultaneously. That obtains a super-resolution model suitable for all kinds of degradation. Kim et al. [5] are the very first to introduce the residual network to the SISR method and propose the VDSR. VDSR uses a deeper network structure to achieve better reconstruction results. Lim et al. [11] build a much deeper network model called EDSR by further improving the residual blocks. Kim et al. [27] introduce a recursive neural network to expand the receptive fields and reduce parameters. Their proposed DRCN employs recursive supervision and skip connection to alleviate gradient explosion and gradient vanishing. Inspired by [41], [42], Li et al. [40] propose to use a recursive neural network together with feedback block and propose SRFBN. Compared with DRCN, SRFBN is more effective in transmitting high-level information back to the previous layer and improving the low-level information coding.

Existing SISR frameworks can be roughly summarized as in Fig. 1(a). Overall, SISR aims to recover SRMS images with rich detail information from their original LRMS versions through specific algorithms. However, since the source images are always degraded, SISR methods are struggled to extract clear detail features from the inputs, so as to reconstruct high-quality SRMS images. Different from previous SISR structures, we propose to take advantage of rich high-frequency information from PAN images. Our designed teacher–student network can precisely separate the high-frequency components from PAN images and transfer them to the super-resolution of LRMS images as supplements.

B. Pansharpening Methods

Pansharpening methods aim to improve the detail information of multispectral images by fusing LRMS images with their corresponding PAN images. According to the specific network structures, existing pansharpening methods can be divided into convolutional neural network-based methods, residual network-based methods, and double-branch structure-based methods. Pansharpening by convolutional neural networks [18], proposed by Giuseppe et al., is the very first pansharpening model based on convolutional neural networks, which also proves the effectiveness of deep learning techniques in pansharpening algorithms compared to traditional approaches. They apply an existing three-layer network SRCNN [26] as a feature extraction part and design their own fusion part. Inspired by the success of [43], Yang et al. [44] propose the PanNet model and introduce the residual module [43] to extract features more effectively. PanNet adopts high-pass filtering strategies on both LRMS and PAN inputs. Edge components are also treated as inputs to improve the generalization ability and robustness of the unseen satellite datasets. Wei et al. [21] also introduce residual learning and build a deep convolutional neural network DRPNN. Their model overcomes the shortcomings of traditional linear models and further improves the fusion performance.

Different from the convolutional neural network-based methods mentioned above, some studies propose a two-stream structure to extract the features from PAN and LRMS inputs, respectively, and then reconstruct the pansharped images according to the fused features. For example, the RSIFNN [45] uses a two-branch network to extract the spatial features from PAN images and the spectral features from LRMS images. TFNet [46] is a dual-flow network that fuses PAN with LRMS images in the feature space. PSGAN [47] improves TFNet by integrating a generative adversarial training strategy. Xu et al. [34] propose a panchromatic sharpening network SDPNet based on surface and deep constraints. To enhance feature representation, they construct two encoder–decoder modules to extract deep and surface features from the image source. They also design dense connection blocks to strengthen feature propagation. The proposed SDPNet model effectively retains the spatial and spectral information from the inputs.

Concisely, those Pansharpening algorithms can be summarized as in Fig. 1(b). Basically, pansharpening methods take both LRMS and PAN images as input to a specific fusion structure and obtain a fusion image with both spectral and

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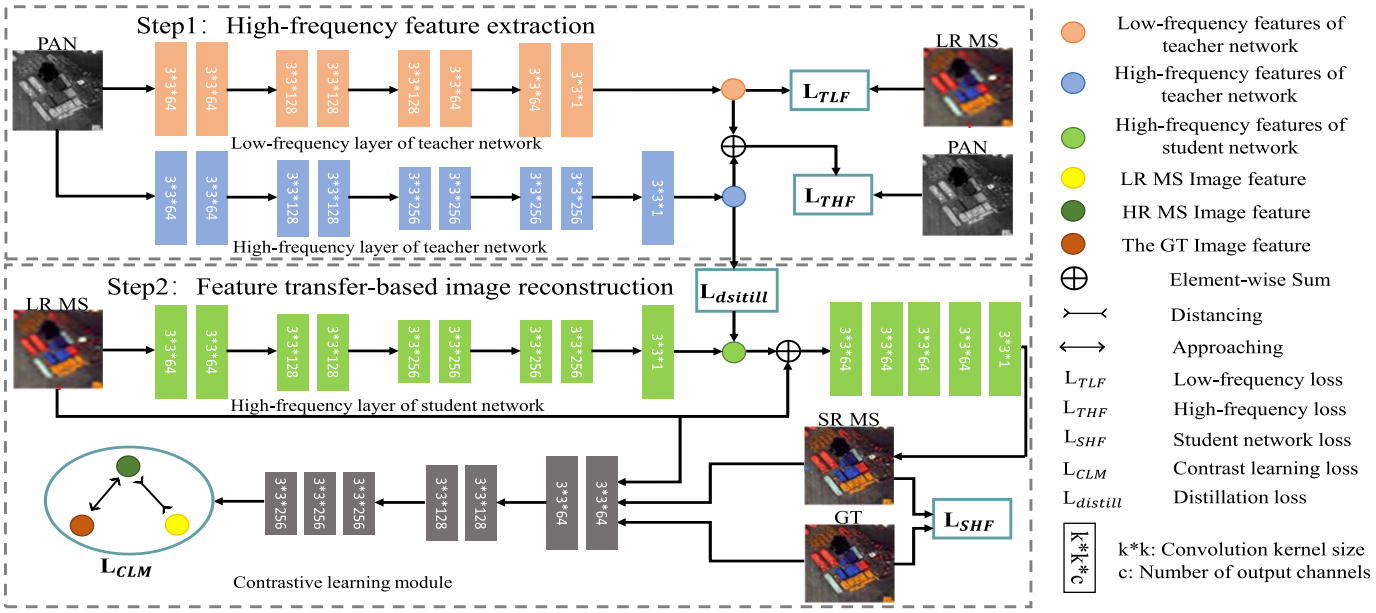


Fig. 2. Our proposed HFFT-PAN framework. First, the features from PAN images are separated by TLF and THF of the teacher network, respectively. Next, the high-frequency components from PAN images are implemented as supervisions to the reconstruction of LRMS images through SHF in the student network. In addition, a contrastive learning mechanism is employed to reinforce the high-frequency details of the super-resolution output.

high-frequency information. However, existing pansharpening methods inevitably demand PAN inputs during testing phases, which significantly limits their applicability under PAN-free circumstances. Instead of fusing PAN and multispectral images straightforward, we design a teacher–student structure to utilize PAN inputs indirectly. High-frequency components extracted from PAN images by the teacher model are considered as supervisions to the reconstruction process of LRMS images within the student model during training. Since the student model is inspired to predict the missing components itself, no auxiliary PAN input is ever required during inference.

III. Proposed Method

A. Motivation

Our goal is to recover and reinforce the high-frequency image details from LRMSs. Based on the observation of the strengths and drawbacks of existing methods, we consider combining the advantages of both pansharpening and SISR methods. In other words, we need PAN images to provide high-frequency information to help the reconstruction, yet no PAN input should be directly involved in the super-resolution process.

Thus, we propose a high-frequency feature transfer for a multispectral image super-resolution network to reach our goal. Unlike most of the existing SISR or pansharpening methods, we introduce a knowledge distillation structure to deal with PAN inputs. In our teacher–student network, the teacher model extracts the high-frequency components from PAN images and transfers them to the multispectral image reconstruction procedure in the student model. The distillation is achieved by a feature supervision paradigm, where the outputs of THF in the teacher model are regarded as high-frequency feature GTs while training SHF in the student model. By doing so, the student model is encouraged to predict the missing high-frequency components based on the LRMS inputs alone. Since no teacher supplement is available while model inference, the proposed method can actually achieve PAN-free testing just like other SISR approaches. Finally, we design a contrastive learning module to further guarantee the quality and fidelity of the generated high-resolution results. The whole pipeline is shown in Fig. 2. We will present this framework specifically in the following order: 1) high-frequency feature learning in Section III-B; 2) feature transfer-based image reconstruction in Section III-C; and 3) optimization in Section III-D.

B. High-Frequency Feature Learning

As shown in Fig. 2, the teacher network consists of two parallel branches THF and TLF, both of which are trained to extract features from PAN images. Specifically, TLF is designed to extract low-frequency components from PAN images, while THF is in charge of the high-frequency part. It should be noticed that what really matters for the reconstruction of LRMS images are the high-frequency components from PAN images, yet it is difficult to collect them straightforward. We find that the original LRMS images already contain most of the low-frequency information. If this can serve as a reference to eliminate the corresponding low-frequency components in PAN images, the rest high-frequency parts will be easy to separate and extract. Thus, we treated the LRMS images as supervision to TLF, forcing the TLF to extract similar frequency information with LRMS inputs. The loss function design of TLF is given as follows:

$$L_{\mathrm{TLF}} = \frac{1}{N}\sum_{n=1}^{N}\left(F_{\mathrm{TLF}}\left(I_{\mathrm{P}}^{n}\right) \ominus I_{\mathrm{M}}^{n}\right) \tag{1}$$

where N indicates the number of a batch, F_{TLF} means the output of TLF, I_P^n stands for the nth PAN image input, I_M^n

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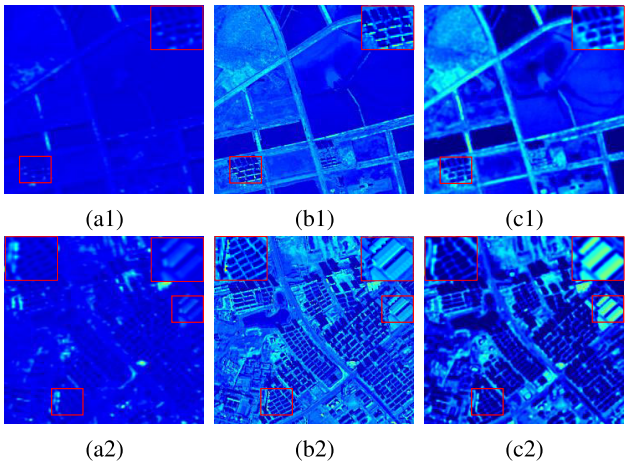


Fig. 3. Visual presentation of feature maps extracted by TLF, THF, and SHF. (a1)–(c1) and (a2)–(c2) Feature output from TLF, THF, and SHF, respectively.

is the nth LRMS image input, and $\ominus$ refers to pixelwise subtraction.

Next, we reconstruct the PAN image by superimposing two-branch feature maps. PAN images are composed of low-frequency features and high-frequency features. Therefore, when we make one branch learn the low-frequency features of PAN images under the supervision of LRMS images, the other branch naturally learns the high-frequency features of PAN images. Therefore, the teacher model is trained to capture features from PAN images as much as possible under PAN self-reconstruction supervision. The loss function for training THF is given as follows:

$$L_{\mathrm{THF}} = \frac{1}{N}\sum_{n=1}^{N}\left(F_{\mathrm{TLF}}\left(I_{\mathrm{P}}^{n}\right) \oplus F_{\mathrm{THF}}\left(I_{\mathrm{P}}^{n}\right) \ominus I_{\mathrm{P}}^{n}\right) \tag{2}$$

where F_{THF} is the output of THF and $\oplus$ means pixelwise addition.

The combination of two losses constitutes the total loss for the teacher model, which can be represented as

$$L_{\mathrm{teacher}} = L_{\mathrm{THF}} + \alpha L_{\mathrm{TLF}} \tag{3}$$

where α is a weight hyperparameter.

To prove that THF effectively learns the ability to separate the high-frequency components from PAN images with the support of TLF, we provide a visualization of feature maps extracted by THF and TLF, as shown in Fig. 3(a1)–(b1) and (a2)–(b2). It is clear that the feature map extracted by THF contains most of the image edge details.

C. Feature Transfer-Based Image Reconstruction

After separating and extracting information from PAN inputs, SHF is designed to accept and apply that knowledge for the reconstruction of LRMS images. In order to prevent the direct utilization of PAN features, we propose to transfer the high-frequency features extracted from THF to SHF through a knowledge distillation mechanism. The high-frequency features collected by THF are treated as pseudolabels to supervise the training of SHF. The distillation loss can be expressed as

$$L_{\mathrm{distill}} = \frac{1}{N}\sum_{n=1}^{n}\left(F_{\mathrm{SHF}}\left(I_{\mathrm{M}}^{n}\right) \ominus f_{\mathrm{THF}}\left(I_{\mathrm{P}}^{n}\right)\right) \tag{4}$$

where F_{SHF} is the output of SHF.

Moreover, to diminish the potential losses of high-frequency information and semantic information in original LRMS images, a residual connection operation is added to SHF. Adding the outputs from SHF with the LRMS source gives the final SRMS results. Similar to the existing SISR methods, we use GT images to supervise SHF training. The loss function of SHF is given as follows:

$$L_{\mathrm{SHF}} = \frac{1}{N}\sum_{n=1}^{N}\left(F_{\mathrm{SHF}}\left(I_{\mathrm{M}}^{n}\right) \oplus I_{\mathrm{M}}^{n} \ominus I_{\mathrm{GT}}^{n}\right) \tag{5}$$

where I_{GT}^{n} is the nth GT image for supervision.

To further improve the fidelity of the generated SRMS results, we introduce an extra contrastive loss to constrain the SHF model. Contrastive learning based on feature representation is a self-supervised learning method [48], [49], [50] that aims to learn a discriminative feature embedding space and a reasonable distance function to measure the similarity between samples. Since contrastive learning is often adopted in high-level visual tasks such as image classification and object detection, it has barely been applied under low-level tasks. We innovatively employ contrastive learning to multispectral image reconstruction tasks. By pushing the generated SRMS results closer to the GT images and pulling them away from LRMS sources at the feature embedding space, the generated SRMS images become similar to the real ones while expanding the differences with LRMS sources. We consider the output of the student model $I_{\hat{\mathrm{M}}}$ and GT image I_{GT} as positive sample pairs while regarding $I_{\hat{\mathrm{M}}}$ and original LRMS input I_M as negative sample pairs. The features of the reconstructed image, GT image, and LRMS image are extracted by the feature extractor. In the feature embedding space, the reconstructed image contains more high-frequency features of the GT image and is far away from the low-frequency features of the LRMS image. The proposed contrastive loss can be formulated as

$$L_{\mathrm{CLM}} = \frac{D\left(G\left(I_{\hat{\mathrm{M}}}\right), G\left(I_{\mathrm{GT}}\right)\right)}{D\left(G\left(I_{\hat{\mathrm{M}}}\right), G\left(I_{\mathrm{M}}\right)\right)} \tag{6}$$

where D() stands for the MAE loss and G() is a feature extractor. We take the pretrained VGG16 network as the basic structure of the feature extractor and implement contrast loss on the feature map outputs of the last convolutional layer in the second module and the third module, respectively. It is worth noting that the input images are all single-channel, so the input channel of the first convolutional layer of the VGG16 network is changed to 1 channel.

The combination of three losses constitutes the total loss for the student model

$$L_{\mathrm{student}} = \lambda L_{\mathrm{distill}} + \beta L_{\mathrm{SHF}} + \gamma L_{\mathrm{CLM}} \tag{7}$$

where β, λ, and γ are weight hyperparameters, respectively.

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As shown in Fig. 3(c1) and (c2), we visualize the feature maps extracted by SHF. It is obvious that the outputs of SHF also contain vivid edge details, which indicates that SHF has effectively learned to predict and recover the high-frequency components under the guidance of THF.

D. Optimization

1) Network Details: The encoder of THF consists of three convolution blocks, and the decoder consists of two convolution blocks. Since TLF is designed to retain the low-frequency detail information of PAN image, the encoder of TLF consists of two convolutional blocks, and the decoder consists of the corresponding two convolutional blocks. To effectively distill high-frequency features to the student network, SHF has the same encoder and decoder as THF. Then, the output SHF high-frequency feature map is superimposed with the LRMS image, and a convolution block containing five convolution layers is added to realize SRMS image reconstruction. Specifically, each of the main branches in our proposed HFFT-PAN adopts modules from VGG16 as basic components. In the teacher model, THF adopts the first three modules from VGG16 while abandoning the pooling layer; TLF takes the first two modules of VGG16 and retains the first pooling but converts the second pooling to a bilinear upsampling layer. In the student model, both SHF and CLM leverage the first three modules of VGG16. It is worth noting that SHF also forsakes the pooling layer, and the output is fused with the source image through another five convolutional layers to obtain the final result. In addition, we convert the LRMS images into HSV color space and separate the V channel as the supervision of the TLF branch. Thus, the output channels of the two teacher networks are the same.

2) Training: To coordinate the proposed method, we introduce a corresponding two-stage training strategy including both teacher and student network training. The teacher model is trained during the first stage. Taking PAN images as inputs, TLF and THF branches are trained by minimizing loss function L_{teacher}, acquiring the ability to separate and extract high-frequency features from PAN images. During the second stage, the LRMS images are input into SHF, and L_{SHF} is used to train the student model, while the parameters of the teacher model remain frozen. The high-frequency knowledge captured from PAN images is transferred to SHF from THF through distillation loss L_{distill}. In addition, we load the pretrained VGG16 parameters as the initialization weights of the contrastive learning module. The contrastive loss L_{CLM} is implemented to guide the reconstruction of the student model. By minimizing the loss function L_{student}, generated SRMS images receive extra texture features while maintaining high fidelity.

Notice that PAN images are only introduced to provide guidance to the reconstruction process during model training. Specifically, the teacher model is only used to supervise the training of the student model so that SHF can learn to predict the high-frequency features similar to what THF extracted from PAN images. During testing, only the student model is required to obtain an SRMS result with a simple LRMS input.

IV. Experiments

A. Implementation

The proposed model is implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU. Adam [51] optimizer is used to optimize the model for 40 epochs. The batch size, initial learning rate, and weight decay are set to 1, 1e-2, and 1e-4, respectively. Our model has parameters of 2.368 M and takes 0.029 s to infer a single-image during testing. While training on 2×, 3×, and 4× downsampled data, we set the hyperparameters (α, β, λ, and γ) in the loss function to (0.1, 2, 0.5, 0.4), (0.05, 1.5, 0.5, 0.4), and (0.001, 1, 0.7, 0.4), respectively.

We construct two datasets for experimentation, SD-1 and SD-2, which originated from the Gaofen-1 dataset[1] and Gaofen-2 dataset,[2] respectively. In the SD-1 dataset, 1587 image pairs (70.1% of all the image pairs) are randomly selected as the training set, and the remaining 680 image pairs (29.9% of all the image pairs) are used as the test set. PAN images in the SD-1 dataset are single-band images with a spatial resolution of 2 m, and the multispectral images are three-band (red, green, and blue) images with a spatial resolution of 8 m. The SD-1 training set has 4761 image pairs, and the test set has 680 image pairs. In the SD-2 dataset, 1844 image pairs (73.1% of all the image pairs) are randomly selected as the training set, and the other 680 image pairs (26.9% of all the image pairs) are selected as the test set. PAN images in the SD-2 dataset are single-band images with a spatial resolution of 1 m, and the multispectral images are three-band (red, green, and blue) images with a spatial resolution of 4 m. The SD-2 training set has 5322 image pairs, and the test set has 680 image pairs. Specifically, the only training set data are augmented by horizontal and vertical flips. Thus, the test and training datasets do not contain overlapping content. Then, the original multispectral images are cropped to 252 × 252, which are treated as GT images. PAN images are also cropped to 252 × 252. The 2×, 3×, and 4× LRMS image datasets are obtained by the bicubic interpolation on GT images. It should also be noted that before training and testing, the LRMS images need to be converted from RGB color space to HSV color space, and the V channel is separated as the LRMS inputs to the proposed HFFT-PAN model.

B. Evaluation Criteria

Five commonly used metrics are adopted to evaluate the quality of reconstructed images and objectively estimate the performance of our method and the competitors. The evaluation metrics include average gradient (AG), entropy (En), mutual information (MI) [52], spatial frequency (SF) [53], and visual information fidelity for fusion (VIFF) [54].

AG can not only be used to evaluate the clarity of fused images but also reflect subtle differences and texture transformation features in the fused images. AG measures the degree of spatial changes from image pixel values. The greater the intensity of pixel value variations, the more pronounced the

[1]https://github.com/zhysora/PSGan-Family
[2]https://github.com/Lihui-Chen/Awesome-Pansharpening

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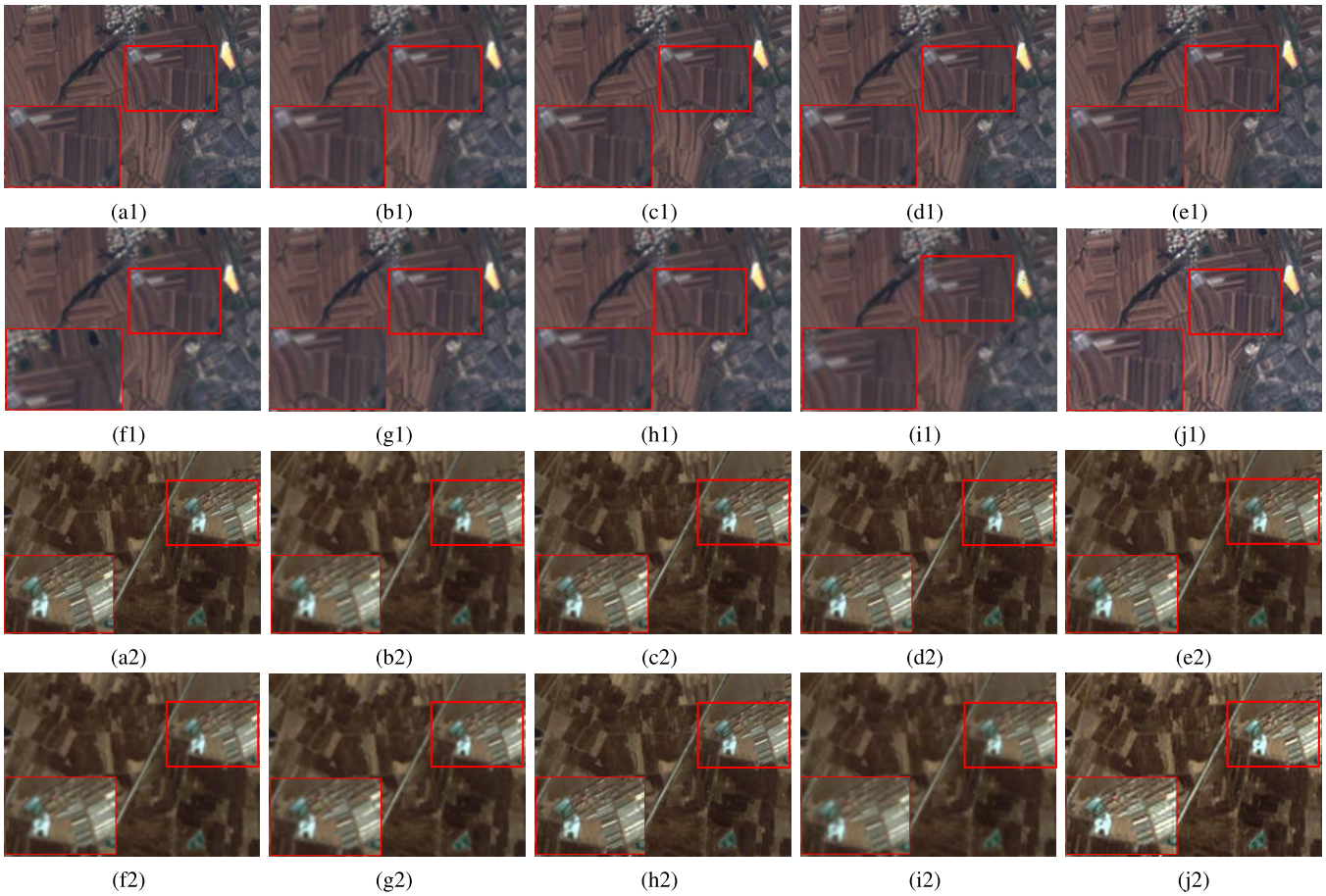


Fig. 4. Visual comparison of our method against existing SISR methods under 2× downsampling on the SD-1 dataset. (a1) and (a2) GT images. (b1) and (b2) LRMS images with 2× downsampling. (c1)–(j1) and (c2)–(j2) Results from FSRCNN, VDSR, SRFBN, SRMD, SRMDnf, RRDB-Linf-LP, SRADSGA, and our method, respectively.

edges and details of the image become, leading to a clearer image. En is an evaluation metric based on the amount of information. The size of En reflects the amount of information carried by an image. The larger the En value of the fused image, the more information it carries. MI can measure the similarity between the fused image and the source image, that is, the amount of information retained through the fusion process. The larger the MI value, the more information is retained in the fused result. SF reflects the details and textures of an image based on the image gradient, which is composed of horizontal frequency and vertical frequency. The higher the SF, the greater the similarity between the source image and the fused image. VIFF is a multiresolution image fusion evaluation algorithm based on visual information fidelity (VIF). VIFF measures the effective visual information of the fused image in each subband through the effective visual information index. The larger the VIFF, the less distortion there is between the fused and source images.

C. Results and Analysis

We compare the results of our proposed method with 11 representative deep learning-based pansharpening methods and SISR methods on SD-1 and SD-2 datasets both quantitatively and qualitatively. Pansharpening competitors include PNN [18], PanNet [44], SDPNet [34], and DRPNN [21], while SISR competitors consist of FSRCNN [8], VDSR [5], SRFBN [40], SRMD [39], RRDB-Linf-LP [55], and SRADSGAN [56].

1) Quantitative Evaluation: We calculate AG, En, MI, SF, and VIFF metrics to compare our methods quantitatively with existing methods on SD-1 and SD-2 datasets.

a) Compared with SISR methods: Tables I and II summarizes the average quality metrics of the proposed method against other SISR methods on SD-1 and SD-2 datasets with 2×, 3×, and 4× downsamplings, respectively. It is clear that our approach achieves the best performance on all metrics. The highest AG, En, and SF values indicate that larger gradients, richer texture, and higher contrast information are retained in our generated results. In addition, the highest MI and VIFF values stand for higher similarity between our results and source images, while more high-frequency information is retained. As shown in Table I, our method outperforms the second best result by 14% of AG, 2% of En, 18% of SF, 2% of MI metric, and 9% of VIFF metric. Especially for LRMS image reconstruction under 4× downsampling, our method is significantly superior to other SISR methods. In fact, this proves the superiority of our goal to enhance the details of multispectral images with PAN images.

In addition, we compare model parameters and inference times with other comparison methods, as shown in Table III.

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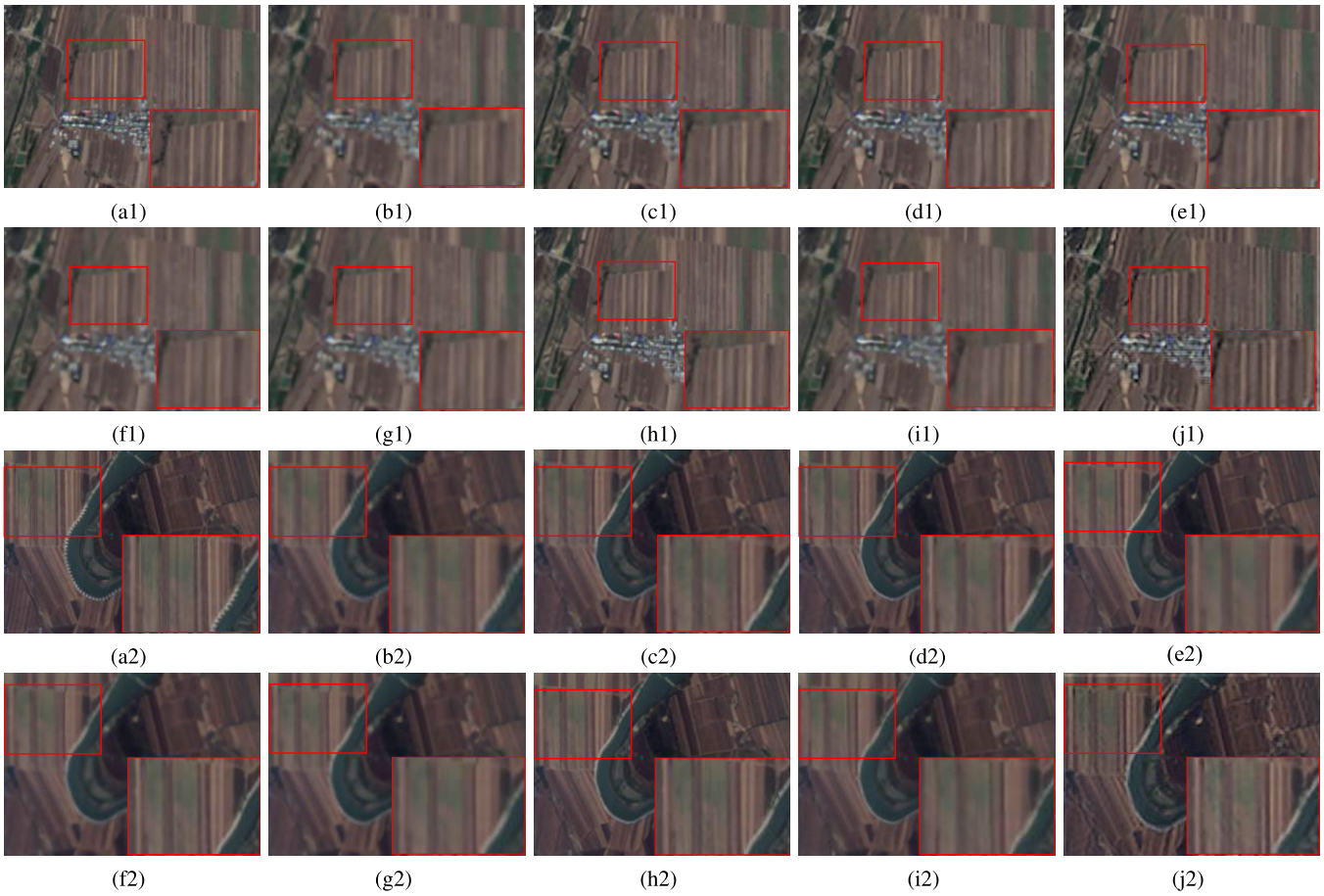


Fig. 5. Visual comparison of our method against existing SISR methods under 3× downsampling on the SD-1 dataset. (a1) and (a2) GT images. (b1) and (b2) LRMS images with 3× downsampling. (c1)–(j1) and (c2)–(j2) Results from FSRCNN, VDSR, SRFBN, SRMD, SRMDnf, RRDB-Linf-LP, SRADSGA, and our method, respectively.

Our method has a faster inference time than SRADSGAN and RRDB-Linf-LP but slower than the other four comparison methods. We will explore lightweight models to increase the speed of inference.

b) Compared with pansharpening methods: Table IV summarizes the average quality metrics of the proposed method against the pansharening methods on the SD-2 dataset. It is obvious that our method achieves the best performance on En, MI, and VIFF metrics and outperforms PNN and SDPNET on AG and SF metrics. The highest En, MI, and VIFF values indicate that more information on the source image is retained in our reconstruction result, and the distortion between the source image and output is minimal. Satisfactory evaluations on AG and SF indicate that our results retain higher gradients, richer texture, and higher contrast. Similar conclusions can be drawn from Table V, which shows that the best performance is achieved by the proposed method on En and MI metrics, and the subbest performance is achieved on AG, SF, and VIFF metrics. Compared to other pansharpening methods, our HFFT-PAN can better retain the information from source images. Although our method does not require PAN inputs during inference, it still achieves competitive results against pansharpening methods.

2) Qualitative Evaluation: Next, we compare the proposed method qualitatively with competitors. As shown in Figs. 4–7, we select eight groups of representative images for visual comparison. In this section, we compare the spectral and spatial features' retention of the output results with the SISR and pansharpening competitors.

Figs. 4–6 show the visual comparison of the 2×, 3×, and 4× downsampling models, respectively. All methods retain the spectral information of source images to some extent. However, compared to other methods, our reconstruction yields superior visual outcomes with fewer artifacts. In particular, the reconstruction results of 4× downsampled images are shown in the rectangular box in Fig. 6(c1)–(j1) and (c2)–(j2). Our results have clearer spatial details, which is intuitively better than other SISR methods.

As shown in Fig. 7(c1)–(g1) and (c2)–(g2), all methods can collect the main components of the source image to a certain extent. Intuitively, our reconstructed image [as shown in the rectangular box of Fig. 7(g1) and (g2)] preserves the spectral features from the multispectral image to the greatest extent and is the closest to the source image. The fusion results of SDPNET [as shown in the rectangular box in Fig. 7(e1) and (e2)] suffer from spectral distortion. The results of DRPNN are shown in the rectangular boxes in Fig. 7(f1) and (f2). Although the edge details are clear, there are spatial artifacts to a certain extent. Overall, our results exhibit more refined spectral features, yet there is still room

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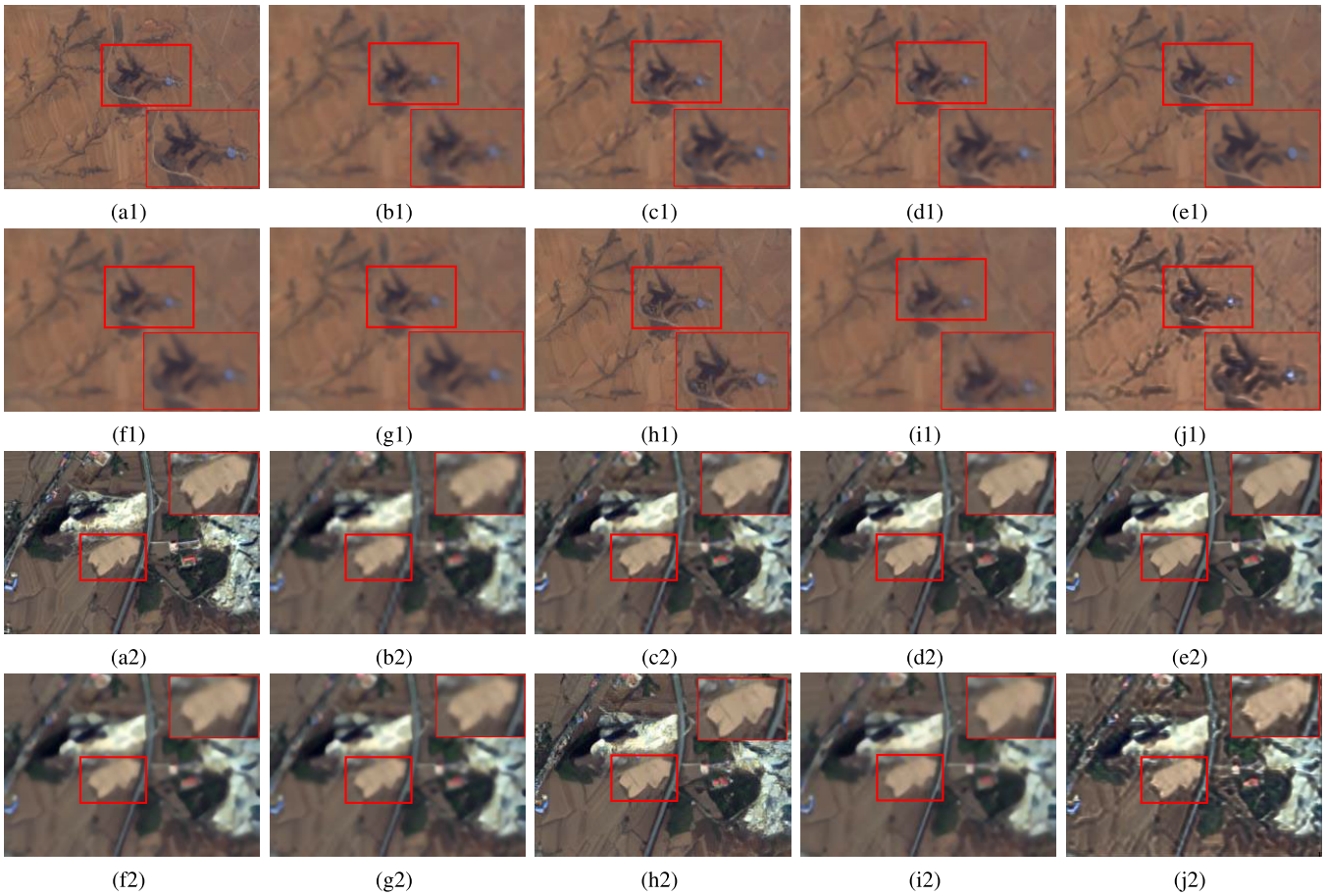


Fig. 6. Visual comparison of our method against existing SISR methods under 4× downsampling on the SD-1 dataset. (a1) and (a2) GT images. (b1) and (b2) LRMS images with 4× downsampling. (c1)–(j1) and (c2)–(j2) Results from FSRCNN, VDSR, SRFBN, SRMD, SRMDnf, RRDB-Linf-LP, SRADSGA, and our method, respectively.

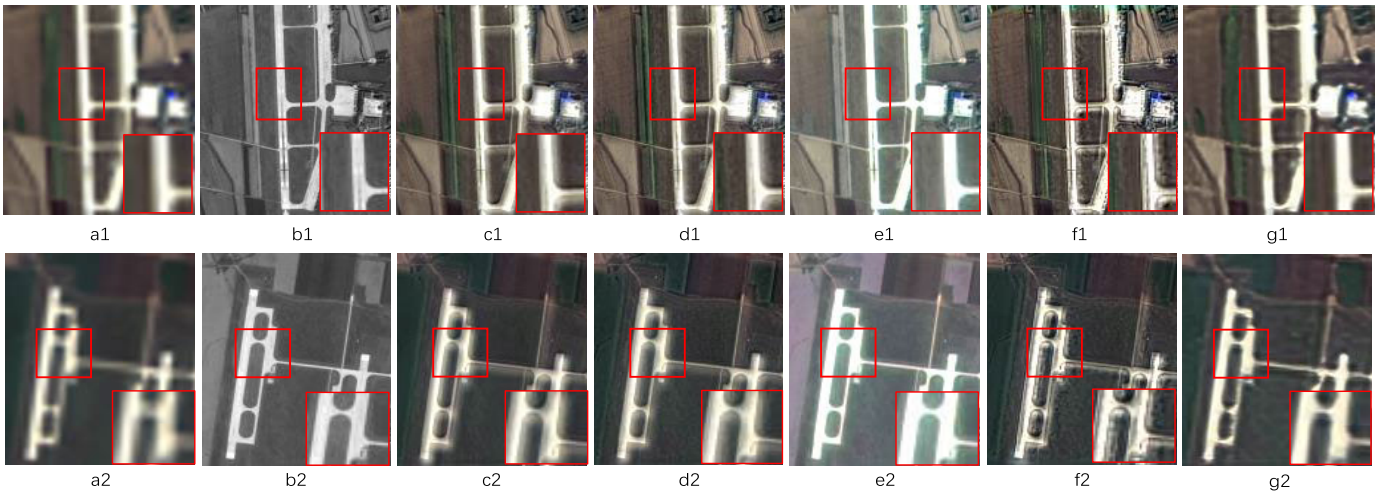


Fig. 7. Visual comparison of our method against existing pansharpening on the SD-1 dataset. (a1)–(g1) and (a2)–(g2) Original PAN images, LRMS images, results of PNN, PanNet, SDPNet, DRPNN, and our model, respectively.

for improvement in spatial details, which is also consistent with our quantitative results.

Through the above quantitative and qualitative comparisons, we are confident in stating that our method has certain advantages in improving the spatial details of multispectral images.

D. Ablation Study

1) Effect of Two-Branch Teacher Network: We design a single-branch PAN image self-reconstruction network as a teacher model for multispectral reconstruction (PSRET). PSRET abandons TLF in the two-branch teacher network and reconstructs PAN images based on the THF branch only,

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TABLE I

AVERAGE QUALITY METRICS OF OUR METHOD AGAINST DIFFERENT SISR METHODS ON THE SD-1 DATASET. THE BEST RESULT IS SHOWN IN RED, AND THE SECOND BEST RESULT IS SHOWN IN **BLUE**

		AG	En	MI	SF	VIFF
2x	FSRCNN	5.700	6.276	12.55	.0454	**.6450**
	VDSR	5.637	6.270	12.54	.0456	.6416
	SRFBN	5.711	6.270	12.54	.0464	.6430
	SRMD	4.368	6.229	12.46	.0342	.6137
	SRMDnf	4.391	6.229	12.46	.0341	.6130
	SRADSGAN	4.507	6.182	12.37	.0349	.6148
	RRDB-linf-LP	**6.188**	**6.277**	**12.56**	**.0491**	.6423
	ours	**7.989**	**6.598**	**13.20**	**.0610**	**.9162**
3x	FSRCNN	4.632	6.249	12.50	.369	.6215
	VDSR	4.570	6.243	12.49	.0377	.6240
	SRFBN	4.724	6.245	12.49	.0396	**.6286**
	SRMD	3.284	6.180	12.36	.0258	.5796
	SRMDnf	3.310	6.182	12.36	.0257	.5807
	SRADSGAN	3.936	6.155	12.31	.0319	.5870
	RRDB-linf-LP	**6.195**	**6.269**	**12.54**	**.0490**	.6250
	ours	**7.099**	**6.454**	**12.91**	**.0579**	**.6863**
4x	FSRCNN	3.794	6.222	12.44	.0302	.6067
	VDSR	3.637	6.205	12.41	.0298	.6024
	SRFBN	3.865	6.213	12.43	.0334	**.6157**
	SRMD	2.581	6.130	12.26	.0202	.5469
	SRMDnf	.2601	.6134	12.27	.0201	.5489
	SRADSGAN	2.980	6.093	12.19	.0224	.5371
	RRDB-linf-LP	**6.326**	**6.263**	**12.53**	**.0490**	.6064
	ours	**9.594**	**6.784**	**13.57**	**.0700**	**.9190**

TABLE II

AVERAGE QUALITY METRICS OF OUR METHOD AGAINST DIFFERENT SISR METHODS ON THE SD-2 DATASET. THE BEST RESULT IS SHOWN IN RED, AND THE SECOND BEST RESULT IS SHOWN IN **BLUE**

		AG	En	MI	SF	VIFF
2x	FSRCNN	1.081	4.608	9.216	.0135	**.5896**
	VDSR	1.834	4.622	9.244	.0139	.5883
	SRFBN	1.732	4.619	9.238	.0136	.5872
	SRMD	1.356	4.575	9.150	.0102	.5792
	SRMDnf	1.574	4.573	9.147	.0102	.5789
	SRADSGAN	1.429	4.523	9.045	.0105	.5875
	RRDB-linf-LP	**1.994**	**4.629**	**9.258**	**.0149**	.5862
	ours	**3.237**	**5.217**	**10.43**	**.0241**	**.8250**
3x	FSRCNN	1.475	4.578	9.156	.0110	.5830
	VDSR	1.493	4.595	9.190	.0114	**.5851**
	SRFBN	1.393	4.585	9.170	.0112	.5844
	SRMD	1.027	4.521	9.042	.0078	.5677
	SRMDnf	1.024	4.520	9.040	.0077	.5679
	SRADSGAN	1.237	4.494	8.988	.0093	.5770
	RRDB-linf-LP	**2.040**	**4.620**	**9.240**	**.0150**	.5807
	ours	**3.175**	**5.309**	**10.62**	**.0304**	**.9386**
4x	FSRCNN	1.237	4.544	9.087	.0092	.5784
	VDSR	1.196	4.546	9.091	.0091	.5768
	SRFBN	1.138	4.549	9.096	.0093	**.5788**
	SRMD	.8163	4.466	8.931	.0062	.5499
	SRMDnf	.8099	4.463	8.925	.0062	.5499
	SRADSGAN	.9367	4.418	8.836	.0072	.5504
	RRDB-linf-LP	**2.050**	**4.607**	**9.215**	**.0149**	.5720
	ours	**4.052**	**5.816**	**11.63**	**.0299**	**1.659**

leaving the other parts unchanged. At this point, since we extract the full features of the PAN image, we distill it behind the skip connection of the SHF network. In addition, we designed a variant of PSRET (PSRET w/o SC) that removes the skip connection in SHF based on PSRET. Quantitative evaluation is shown in Table VI. Compared with HFFT-PAN, the reconstruction effect of PSRET and PSRET w/o SC is poor, and all indicators are decreased. AG increases by 18%, En increases by 0.8%, and SF increases by 15%, which indicates that HFFT-PAN has richer edge details and higher contrast. In addition, MI increases by 0.8% and VIFF increases by 21%, which indicates that the reconstructed image of HFFT-PAN has higher similarity to the source image. The visual comparison results are shown in Fig. 8. Compared with HFFT-PAN, PSRNET and PSRNET w/o SC retain less high-frequency details.

2) Effect of High-Frequency Feature Transfer Based on Distillation: As described in Sections III-B and III-C, a teacher–student structure is implemented to transfer the high-frequency features of PAN images to LRMS reconstruction through knowledge distillation. Here, we analyze the role of distillation design in our HFFT-PAN model. The quantitative evaluation of the SD-1 dataset is shown in Table VI. Compared to HFFT-PAN without distillation (HFFT-PAN w/o DL), the reconstructed image of HFFT-PAN is improved in terms of edge details, resulting in a 59% increase in AG, an 8% increase in En, and a 58% increase in SF. In addition, the reconstructed image of HFFT-PAN has a higher similarity to the source image, causing an 8% increase in MI and an 82% increase in VIFF. We also visually compare the reconstruction results of HFFT-PAN w/o DL and HFFT-PAN, as shown in Fig. 8(d1), (d2), (e1), and (e2). Fig. 8(d1) and (d2) presents the poor edge details, while Fig. 8(e1) and (e2) has sharper edge details.

3) Effect of Self-Supervision Based on Contrastive Learning: As described in Section III-C, we introduce an extra CLM to the student model to improve the fidelity of the reconstructed images. Here, we demonstrate the role of the contrast learning module in HFFT-PAN. Quantitative evaluation is shown in Table VI. Compared with HFFT-PAN without CLM (HFFT-PAN w/o CLM), HFFT-PAN refines the image contrast (e.g., AG, En, and SF) and fidelity (e.g., MI and VIFF). We can see that all evaluation metrics receive an increase of at least 5%. Fig. 8(c1), (c2), (e1), and (e2) shows the visual comparison of HFFT-PAN w/o CLM and HFFT-PAN models, respectively. It is clear that Fig. 8(e1) and (e2) has a higher edge fidelity than HFFT-PAN w/o CLM.

It is obvious that the reconstructed images of HFFT-PAN contain abundant edge detail information and have higher similarity to the source images, which indicates that the two-branch teacher model, distillation, and CLM all make

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TABLE III
COMPARISON OF MODEL PARAMETERS AND INFERENCE TIMES OF SISR METHODS

	FSRCNN	VDSR	SRMD	SRMDnf	SRFBN	SRADSGAN	RRDB-Linf-LP	Ours
Parameter/M	.013	.665	1.553	1.552	3.463	11.069	22.100	2.368
Inference time/s	.010	.012	.022	.023	.027	.040	.123	.029

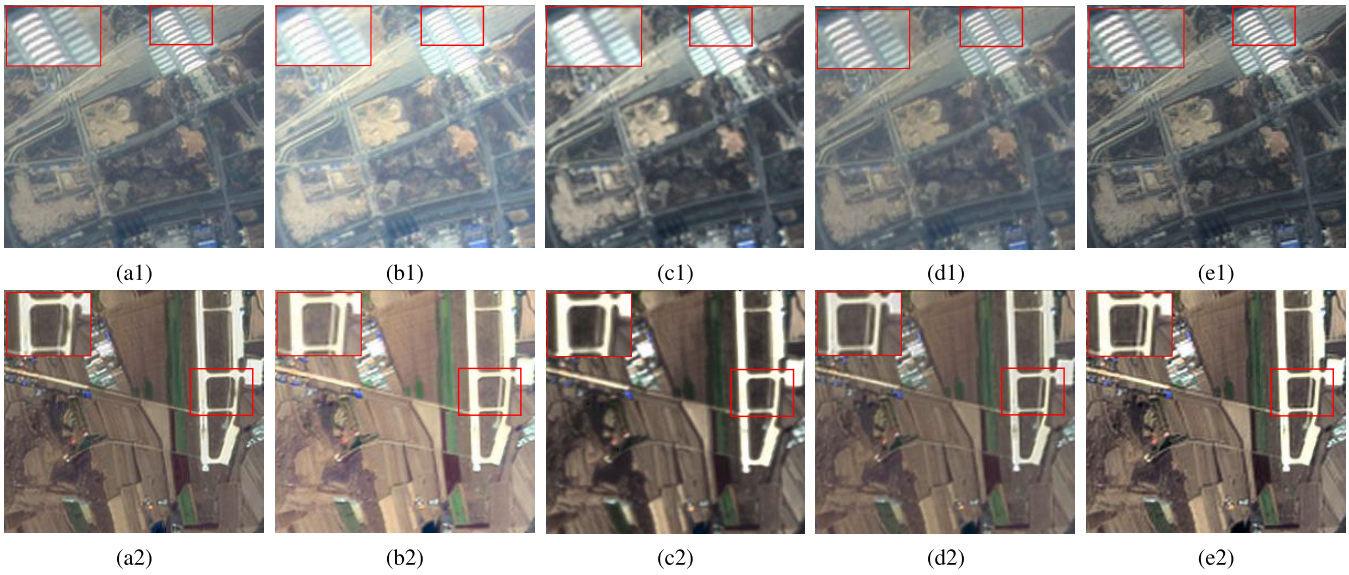


Fig. 8. Visual comparison results of different teacher models, noncontrast learning modules, and nondistillation mechanisms on the SD-1 dataset. (a1) and (a2) PSRNET. (b1) and (b2) PSRNET w/o SC. (c1) and (c2) HFFT-PAN w/o CLM. (d1) and (d2) HFFT-PAN w/o DL. (e1) and (e2) Ours.

TABLE IV
AVERAGE QUALITY METRICS OF OUR METHOD AGAINST DIFFERENT PANSHARPENING METHODS ON THE SD-2 DATASET. THE BEST RESULT IS SHOWN IN RED, AND THE SECOND BEST RESULT IS SHOWN IN **BLUE**

	AG	En	MI	SF	VIFF
PNN	3.487	4.895	9.789	.0263	.9413
PanNet	**4.609**	5.078	10.16	**.0335**	1.104
SDPNet	2.576	4.957	9.914	.0198	1.028
DRPNN	**6.106**	**5.300**	**10.60**	**.0442**	**1.256**
Ours	4.052	**5.816**	**11.63**	.0299	**1.659**

TABLE V
AVERAGE QUALITY METRICS OF OUR METHOD AGAINST DIFFERENT PANSHARPENING METHODS ON THE SD-1 DATASET. THE BEST RESULT IS SHOWN IN RED, AND THE SECOND BEST RESULT IS SHOWN IN **BLUE**

	AG	En	MI	SF	VIFF
PNN	6.716	6.241	12.48	.0511	.5793
PanNet	8.775	6.319	12.64	.0641	.6304
SDPNet	8.506	**6.478**	12.96	.0661	**.9968**
DRPNN	**11.29**	6.413	12.83	**.0827**	.6277
Ours	**9.594**	**6.784**	**13.57**	**.0700**	**.9190**

great contributions to reinforce the high-frequency components in SRMS results and guarantee high image fidelity.

4) Influence of Parameter Values: There are parameters α, β, λ, and γ involved in this model. Here, we illustrate the choice of parameters. First, we demonstrate the selection of

TABLE VI
EVALUATION RESULTS OF FEATURE EXTRACTION MODEL, CONTRASTIVE LEARNING MODULE, AND DISTILLATION IN HFFT-PAN ON THE SD-1 DATASET. THE BEST RESULT IS SHOWN IN RED

	AG	En	MI	SF	VIFF
PSRNET	6.265	6.130	12.26	.0511	.5239
PSRNET w/o SC	6.770	6.546	13.09	.0530	.7532
HFFT-PAN w/o CLM	5.306	6.244	12.49	.0508	.6822
HFFT-PAN w/o DL	5.023	6.109	12.22	.0385	.5025
Ours	**7.989**	**6.598**	**13.20**	**.0610**	**.9162**

hyperparameter α in the teacher network. We quantitatively compare the high-frequency characteristic graphs of THF output under different α parameters, and the results are shown in Table VII. It is obvious that the best results are achieved when α is 0.1, 0.05, and 0.001, respectively, on the $2\times$, $3\times$, and $4\times$ datasets.

Next, based on the optimal value of α, we discuss the hyperparameters β, λ, and γ in the student model. Here, we present the detailed process of hyperparameter selection on the $2\times$ dataset, and the quantitative evaluation is shown in Table VIII. When β, λ, and γ are 2, 0.5, and 0.4, respectively, the results of each index reach the best. We also give the quantitative results of three hyperparameters on the $3\times$ and $4\times$ datasets under different values in Table IX. The best results are obtained when β, λ, and γ are 1.5, 0.5, and 0.4 and 1, 0.7, and 0.4 on the datasets $3\times$ and $4\times$, respectively.

Based on the above discussion on the selection of hyperparameters, α and β decrease with the increase in the

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TABLE VII

QUANTITATIVE RESULTS OF α PARAMETER SELECTION ON THE SD-1 DATASET AT DIFFERENT DOWNSAMPLING MULTIPLES. THE BEST RESULT IS SHOWN IN RED

	2x				3x				4x			
α	0.5	0.1	0.05	0.01	0.1	0.05	0.01	0.005	0.05	0.01	0.005	0.001
AG	4.940	**7.344**	6.141	6.253	3.010	**8.071**	4.841	6.617	6.846	6.185	8.349	**9.015**
En	4.982	**5.362**	5.224	5.239	4.279	**5.238**	4.529	4.554	5.718	5.323	5.097	**5.778**
MI	9.963	**10.72**	10.45	10.48	8.558	**10.48**	9.059	9.108	11.44	10.65	10.19	**11.56**
SF	.0396	**.0620**	.0492	.0508	.0239	**.0693**	.0422	.0583	.0589	.0504	.0705	**.0743**
VIFF	.4459	**.4549**	.4471	.3549	.2981	**.8295**	.4195	.2274	.7046	.4584	.9148	**.9710**

TABLE VIII

QUANTITATIVE RESULTS OF β, λ, AND γ PARAMETERS' SELECTION ON THE SD-1 DATASET AT 2× DOWNSAMPLING. THE BEST RESULT IS SHOWN IN RED

β	0.4							1.0							2.0						
λ	0.0	0.5	1.0	1.5	1.0			0.2	0.5	0.8	1.2	0.5			0.2	0.5	1.0	1.5	0.5		
γ	0.0				0.5	1.0	1.5	0.0				0.4	0.8	1.2	0.1				0.4	0.8	1.2
AG	2.259	2.949	4.126	2.996	6.758	5.773	6.209	4.450	6.001	5.143	5.432	6.818	6.923	5.242	4.906	7.189	6.147	6.168	**7.989**	6.049	7.186
En	5.416	5.459	5.963	5.874	6.243	6.108	5.240	5.748	6.374	6.123	5.653	5.854	6.273	6.117	6.009	6.395	6.423	6.179	**6.598**	6.261	6.273
MI	10.83	10.92	11.93	11.75	12.49	12.22	10.48	11.50	12.75	12.25	11.31	11.71	12.55	12.23	12.02	12.79	12.85	12.36	**13.20**	12.52	12.55
SF	.0170	.0252	.0347	.0315	.0508	.0463	.0541	.0372	.0555	.0457	.0567	.0566	.0608	.0469	.0426	.0564	.0605	.0546	**.0610**	.0464	.0559
VIFF	.2337	.2885	.5269	.1867	.5579	.4847	.4951	.3299	.8705	.5764	.5066	.3375	.6619	.4936	.4184	.5642	.4959	.4461	**.9162**	.6273	.6663

TABLE IX

QUANTITATIVE RESULTS OF β, λ, AND γ PARAMETERS' SELECTION ON THE SD-1 DATASET AT 3× AND 4× DOWNSAMPLING. THE BEST RESULT IS SHOWN IN RED

	3x				4x			
$\beta/\lambda/\gamma$	2/0.5/0.4	1.5/0.5/0.4	1.5/0.7/0.4	1.5/0.5/0.6	1.5/0.5/0.4	1/0.5/0.4	1/0.7/0.4	1/0.7/0.2
AG	5.046	**7.099**	6.605	5.642	3.587	6.392	**9.594**	4.634
En	5.759	**6.454**	6.244	6.283	5.718	6.359	**6.784**	5.674
MI	11.52	**12.91**	12.49	12.57	11.44	12.72	**13.57**	11.35
SF	.0560	**.0579**	.0467	.0432	.0273	.0508	**.0700**	.0345
VIFF	.3410	**.6863**	.6348	.5840	.3261	.6913	**.9190**	.3076

subsampling multiple, λ tends to increase with the increase in the subsampling multiple, and γ of 0.4 is ideal.

α in (3) balances the two loss functions L_{THF} and L_{TLF} of the teacher network during training. A smaller α value allows THF to better learn the high-frequency features of PAN images. β in (7) has an impact on the super-resolution reconstruction results of LRMS images, and a larger β value can make the reconstruction results more close to GT images. λ in (7) is the weight coefficient of distillation loss, and a larger λ value can enable SHF to better learn the high-frequency features of PAN images. γ in (7) is the weight coefficient of the relative learning loss, and a larger γ value can improve the fidelity of SRMS images.

V. CONCLUSION

In this article, we propose a high-frequency feature transfer-based method HFFT-PAN for multispectral image super-resolution reconstruction to improve the high-frequency information in multispectral images. A distillation mechanism is introduced to transfer the rich high-frequency features contained in PAN images to the reconstruction of LRMS images so that the reconstructed images can possess rich high-frequency details. In addition, a contrastive learning strategy is employed to enhance the high-frequency and texture features of SRMS images under a self-supervision manner. Both qualitative and quantitative experiments show that compared with existing pansharpening methods, our proposed HFFT-PAN can achieve competitive results without relying on any PAN inputs. Compared with existing SISR methods, our method shows better performances in recovering the high-frequency details in multispectral images. Especially for the reconstruction of 4× downsampled LRMS images, the reconstructed image of HFFT-PAN is remarkably better than other SISR methods, which indicates that our HFFT-PAN has significant advantages in dealing with LRMS image reconstruction tasks. Although the proposed method does not need PAN images during testing, it is inevitable to rely on PAN images during the training phase. Therefore, we will research the generation of high-frequency features to avoid dependence on PAN images during the training phase.

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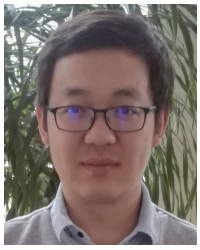
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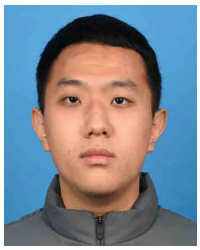
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